

Let $f(x) = \frac{\alpha x}{x+1}$, $x \neq -1$ then for what value of α is

$$f[f(x)] = x.$$

A. $\sqrt{2}$

B. $-\sqrt{2}$

C. 1

D. -1

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Consider the function such that

$$f(x) = \frac{\alpha x}{x+1}, x \neq -1$$

Compute $f[f(x)]$

$$\begin{aligned} f[f(x)] &= f\left[\frac{\alpha x}{x+1}\right] \\ &= \frac{\alpha\left(\frac{\alpha x}{x+1}\right)}{\left(\frac{\alpha x}{x+1}\right)+1} \\ &= \frac{\alpha^2 x}{\alpha x + x + 1} \\ &= \frac{\alpha^2 x}{(\alpha + 1)x + 1} \end{aligned}$$

Given that $f[f(x)] = x$ thus,

$$\frac{\alpha^2 x}{(\alpha + 1)x + 1} = x \Rightarrow \alpha^2 = (\alpha + 1)x + 1; x \neq 0$$

Thus,

$$\begin{aligned} \alpha^2 - (\alpha + 1)x - 1 &= 0 \\ \alpha^2 - 1 - (\alpha + 1)x &= 0 \\ (\alpha + 1)(\alpha - 1) - (\alpha + 1)x &= 0 \\ (\alpha + 1)(\alpha - 1 - x) &= 0 \end{aligned}$$

Hence,

$$\alpha + 1 = 0 \text{ or } \alpha - 1 - x = 0$$

As $f[f(x)] = x$ for all x so, we consider $\alpha + 1 = 0$ that is $\alpha = -1$

Thus, the **OPTION D** is the correct.